Lab 2: Diode Limiting Circuits

ECE 0102: Microelectronic Circuits + Lab

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Team Roles:

Activity	**Student Name**
Circuit Construction	Stefan Zhang
Data Collection	Knight Liu
Data Analysis	Stefan Zhang; Knight Liu
Answers to Questions	Stefan Zhang; Knight Liu
Lab Report Writing	Stefan Zhang

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1) Objectives and Background

The objective of this lab was to study diode limiting circuits and protection circuits using the constant-voltage-drop model. In this lab-2, several limiter circuits were analyzed, designed, and tested in both the time domain and the transfer-characteristic view. The input and output waveforms were observed on the oscilloscope, and the X-Y mode was used to display the transfer characteristics directly.

A silicon diode can be approximated by the constant-voltage-drop model in circuit analysis [1]. In this model, the diode is treated as an open circuit when it is off, and as a constant voltage drop of about 0.7 V when it is on. Based on this model, the clipping levels of the limiter circuits can be predicted before the experiment.

2) Equipment Used

Table 1: Equipments Used in Lab 1

Item	**Quantity**
1N4007 Silicon [4]	2–3
Resistor (10 kΩ)	1–2
Function Generator	1
Oscilloscope	1
DC Power Supply / Bias Source	1
Breadboard and Connecting Wires	1 set

3) Action 1.1: Analysis and Measurement of the Circuit in Figure 1

3.1) Circuit Analysis

Figure from Action 1.1 is a biased diode limiter[2]. Using the constant-voltage-drop model, the diode turns on only when the output voltage attempts to rise above the bias level plus the diode drop. Since the bias source is 3 V, the conduction threshold is [3]

$$V_{\text{clip}} = 3 + 0.7 = 3.7\,\text{V}$$

Therefore, the transfer characteristics can be written as

$$V_{OUT} = \begin{cases} V_{IN}, & V_{IN} \leq 3.7\,\text{V} \\ 3.7\,\text{V}, & V_{IN} > 3.7\,\text{V} \end{cases}$$

This means the positive part of the output is limited to about 3.7 V, while the rest of the signal follows the input.

3.2) Analyzed Plots

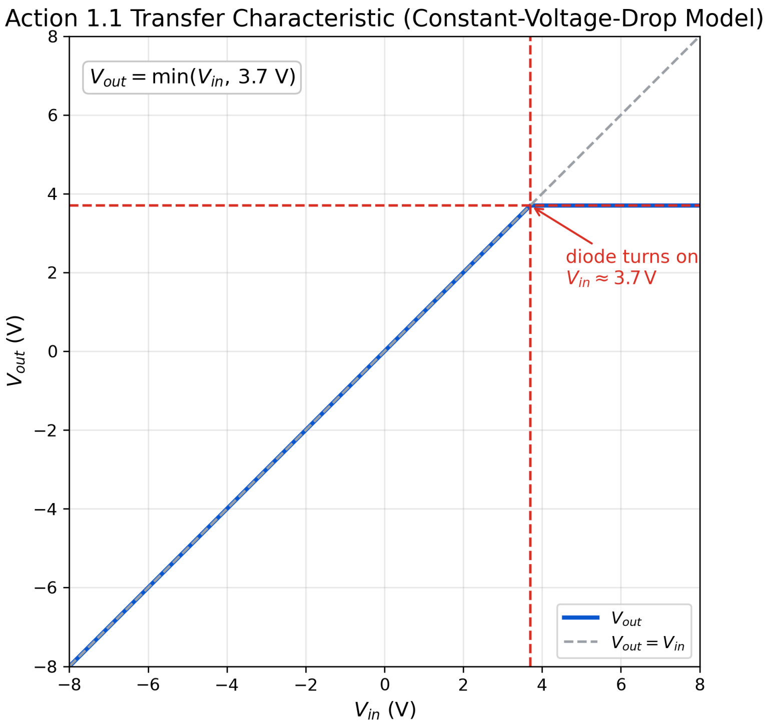

Figure 1: Analyzed transfer characteristics for Action 1.1.

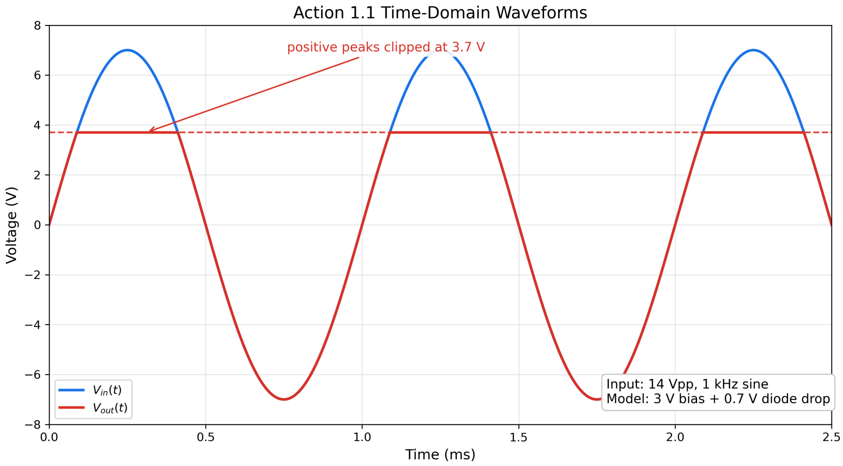

Figure 2: Analyzed input and output waveforms for Action 1.1.

3.3) Procedure

- The circuit was assembled on the breadboard.
- A sinusoidal input signal with

$$V_{IN} = 14\,\mathrm{V_{PP}}, \qquad f = 1\,\mathrm{kHz}$$

  was applied from the function generator.
- The oscilloscope was used to display $V_{IN}(t)$ and $V_{OUT}(t)$ at the same time.
- Then, the oscilloscope was switched to X-Y mode to display the transfer characteristics of the circuit.

3.4) Experimental Setup

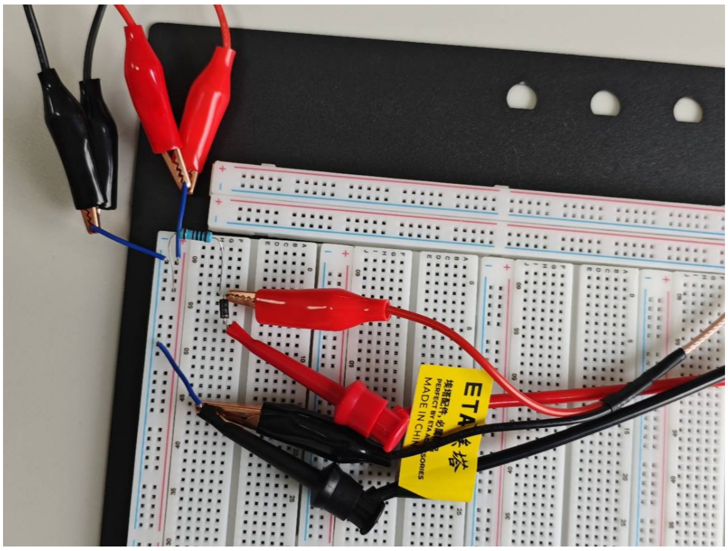

Figure 3: Experimental setup for Action 1.1.

3.5) Measured Waveforms and Discussion

Figure 4: Measured input and output waveforms for Action 1.1.

The measured output waveform should follow the input signal when the diode is off. When the input becomes large enough to make the output exceed about 3.7 V, the diode turns on and limits the output voltage near this level. As a result, the positive peak is clipped, while the negative half-cycle is almost unchanged.

In the actual screenshot, compare the measured clipping level with the theoretical value of about 3.7 V. Any small difference can be explained by diode non-ideality, source error, and measurement uncertainty.

3.6) Measured Transfer Characteristics and Discussion

Figure 5: Measured transfer characteristics for Action 1.1.

The X-Y plot should show an approximately linear region with slope close to 1 before clipping, followed by a nearly horizontal region after the diode turns on. This agrees with the constant-voltage-drop model. In the final report, describe whether the turning point in the measured curve is close to the predicted value.

4) Action 1.2: Design of a Limiter with a Lower Clipping Level of −2.7 V

4.1) Designed Circuit and Operation

To obtain the transfer characteristics in Figure 2, the circuit should keep the output equal to the input when the signal is above the lower limit, and clamp the output near −2.7 V when the input becomes more negative.

Using the constant-voltage-drop model, the required clipping level is

$$V_{\text{clip}} = -(2.0 + 0.7) = -2.7\,\text{V}$$

Thus, a suitable design is a biased negative limiter composed of a resistor and a diode in series with a 2 V bias source, with the diode orientation chosen so that it conducts when V_{OUT} attempts to go below −2.7 V.

The transfer characteristics can be expressed as

$$V_{OUT} = \begin{cases} -2.7\,\text{V}, & V_{IN} < -2.7\,\text{V} \\ V_{IN}, & V_{IN} \geq -2.7\,\text{V} \end{cases}$$

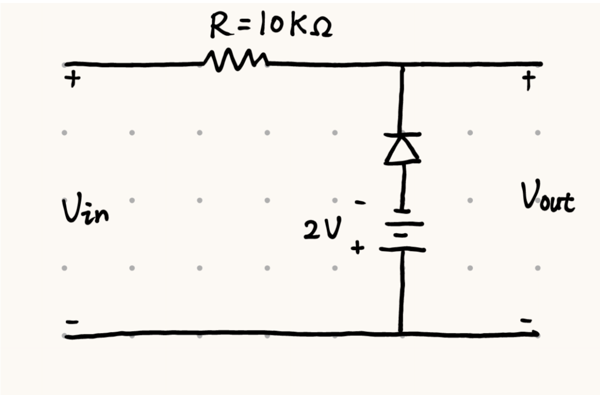

Figure 6: Designed circuit schematic for Action 1.2.

4.2) Procedure

- The designed circuit was assembled on the breadboard
- The designed circuit was tested using a sinusoidal input signal with

$$V_{IN} = 14\,\text{V}_{\text{PP}}, \qquad f = 1\,\text{kHz}$$

- The oscilloscope was used first in normal mode to observe $V_{IN}(t)$ and $V_{OUT}(t)$
- Then in X-Y mode to display the transfer characteristics.

4.3) Experimental Setup

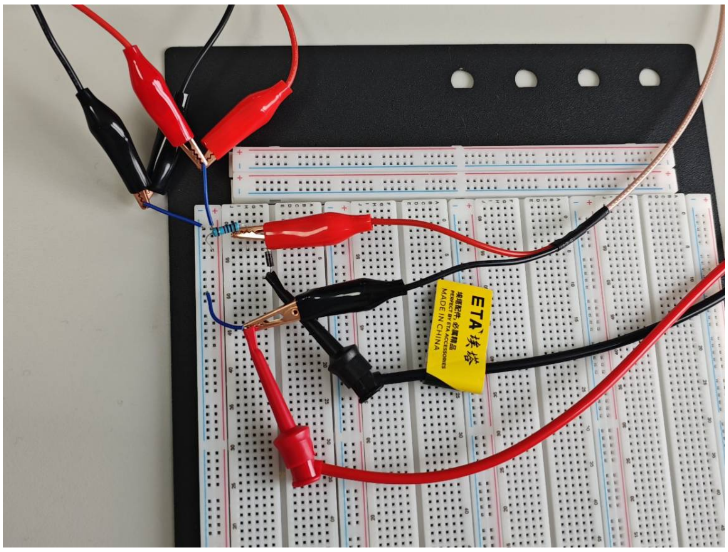

Figure 7: Experimental setup for Action 1.2.

4.4) Measured Waveforms and Discussion

Figure 8: Measured input and output waveforms for Action 1.2.

The measured waveform should show that the output follows the input over most of the range, but it stops decreasing once it reaches about $-2.7\,\mathrm{V}$. In other words, the negative peak is clipped, while the positive part is almost unchanged.

In the final version of the report, compare the measured lower clipping level with the designed value and note any deviation.

4.5) Measured Transfer Characteristics and Discussion

Figure 9: Measured transfer characteristics for Action 1.2.

The measured X-Y curve should contain a horizontal segment near $-2.7\,\mathrm{V}$ and a linear segment with slope close to 1 above this level. This is the expected behavior of a lower limiter.

5) Action 1.3: Design of a Double Limiter with Levels at $-1\,\mathrm{V}$ and $3\,\mathrm{V}$

5.1) Designed Circuit and Operation

To realize the transfer characteristics in Figure 3, the circuit must limit the output at both the lower side and the upper side. Between the two clipping levels, the output should follow the input directly.

A suitable design is a double limiter formed by two oppositely oriented diode branches with different bias sources. One branch turns on when the output tries to rise above about $3\,\mathrm{V}$, and the other branch turns on when the output tries to fall below about $-1\,\mathrm{V}$.

Thus, the transfer characteristics can be written as

$$V_{OUT} = \begin{cases} -1\,\mathrm{V}, & V_{IN} < -1\,\mathrm{V} \\ V_{IN}, & -1\,\mathrm{V} \le V_{IN} \le 3\,\mathrm{V} \\ 3\,\mathrm{V}, & V_{IN} > 3\,\mathrm{V} \end{cases}$$

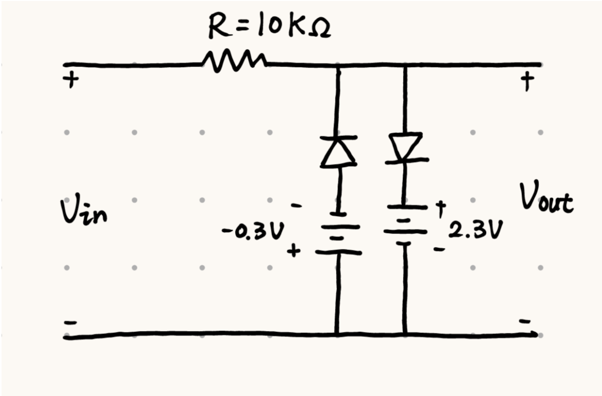

Figure 10: Designed circuit schematic for Action 1.3.

5.2) Procedure

- The designed circuit was assembled on the breadboard
- The designed circuit was tested using a sinusoidal input signal with

$$V_{IN} = 14\,\mathrm{V_{PP}}, \qquad f = 1\,\mathrm{kHz}$$

- The oscilloscope was used to observe the time-domain waveforms
- Then the X-Y transfer characteristics

5.3) Experimental Setup

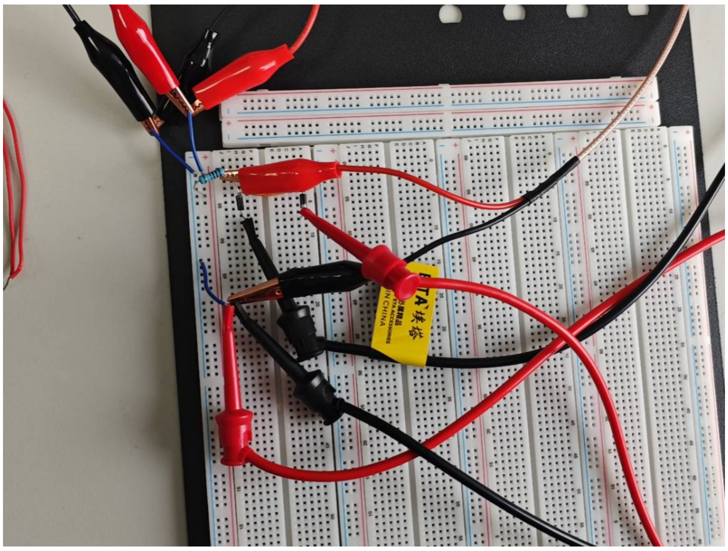

Figure 11: Experimental setup for Action 1.3.

5.4) Measured Waveforms and Discussion

Figure 12: Measured input and output waveforms for Action 1.3.

The measured output waveform should follow the input only in the middle range. When the input becomes too positive, the output is limited near 3 V. When the input becomes too negative, the output is limited near −1 V. Therefore, both the upper and lower parts of the waveform are clipped.

In the final report, use your screenshot to check whether the measured clipping levels are close to the designed values.

5.5) Measured Transfer Characteristics and Discussion

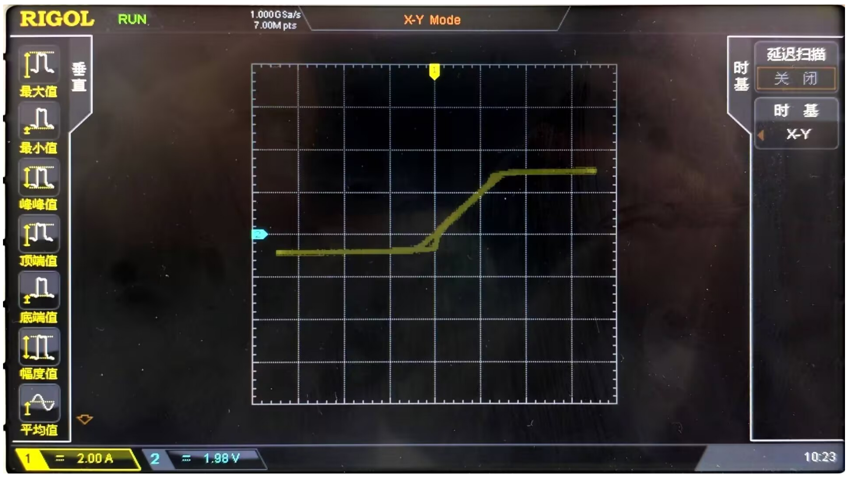

Figure 13: Measured transfer characteristics for Action 1.3.

The X-Y plot should show three regions: a lower horizontal region near −1 V, a central linear region with slope close to 1, and an upper horizontal region near 3 V. This confirms the operation of the double limiter circuit.

6) Conclusion

In this lab, diode limiting circuits were analyzed and tested using the constant-voltage-drop model. The experimental results should show that the diode can be used to restrict the output voltage to a desired range. In Action 1.1, the circuit worked as an upper limiter. In Action 1.2, a lower limiter was designed and tested. In Action 1.3, a double limiter was designed to clip both the positive and negative parts of the signal. The measured waveforms and X-Y plots can be compared with the theoretical predictions to evaluate the accuracy of the design.

References

[1] Microelectronic Circuits (8th Edition), Sedra, Smith, Carusone, Gaudet, Oxford University Press.

[2] SCUPI 2026S: Microelectronics Circuits, *Lab #2 Assignment: Diode Limiting Circuits*, course handout, 2026.

[3] SCUPI 2026S: Microelectronics Circuits, *Lecture 2: Diode Circuits*, course slides, 2026.

[4] Micro Commercial Components, *1N4001–1N4007 General Purpose Rectifiers Datasheet*, accessed for diode model reference.